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A machine learning framework to predict kidney graft failure with class imbalance using Red Deer algorithm

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ABSTRACT

The prevalence of chronic kidney diseases has raised the demand for kidney replacement therapy over recent years. Although kidney transplantation provides better life quality than other alternatives, the replacement might face rejection and endanger individuals' lives. Therefore, predicting this rejection is of great importance, and due to the complexity of the human body's immune system, it is a complicated task. Machine learning has been widely used in disease diagnoses such as graft rejection. One of the common issues in disease diagnosis is the class imbalance problem. The predictive model is likely to misclassify all of the minority class samples in such cases. This paper proposes a method that can accurately predict kidney rejection in an imbalanced dataset. We have adopted kidney transplantation data from 378 patients collected from 1994 to 2011. Our main contribution is to develop a novel clustering method using Red Deer Algorithm (RDA), which is later utilized in proposing a three-stage clustering-based undersampling approach to handle the class imbalance problem so that the final predictive models can classify the given data more accurately. The undersampling method includes denoising, RDA clustering, and sample selection. Moreover, the proposed clustering method is also used to reduce the data dimensionality. Subsequently, five different classification algorithms such as Support Vector Machine (SVM), Artificial Neural Network (ANN), K-Nearest Neighbor (KNN), Decision Tree (DT), and an ensemble method are used to predict kidney graft rejection. Then, these classifiers are compared in terms of five performance evaluation metrics: accuracy, sensitivity, specificity, *F1* score, and area under the Receiver Operating Characteristic (ROC) curve (AUC). The obtained results indicate that the decision tree model outperformed other algorithms and achieved 0.96, 0.94, 0.97, 0.95, and 0.95 for accuracy, sensitivity, specificity, *F1* score, and AUC, respectively. Hence, it is highly recommended to use the proposed method as a decision support system for clinical experts to predict kidney transplantation failure.

1. Introduction

Due to the prevalence of Chronic Kidney Diseases (CKD), most of which lead to end-stage of kidney disease, the demand for kidney replacement therapy (KRT) has increased in recent decades (Wang et al., 2018). Although there have always been different ways to deal with KRT, like dialysis, kidney transplantation gives better life quality and survival rates compared to other alternatives (Karam et al., 2003). However, it is difficult to respond to all kidney graft demands immediately and efficiently for two main reasons (Cecka & Gritsch, 2008). First of all, the demand for kidney grafts is far more than the number of kidney donors, resulting in long waiting queues for kidney transplantation. Secondly, there is always a possibility that the recipient body does not accept the transplanted organ, so the transplantation faces failure.

Regarding these difficulties, predicting graft failure becomes essential in organ allocation because this prediction can decrease the rate of unsuccessful transplantation and minimize the number of people who have to return to the already-burdened waiting list. In this situation, models that are able to predict the graft outcome effectively can help professionals decide which organ should be allocated to whom recipient.

Machine learning (ML) is a set of computer algorithms that can learn from patterns in the underlying data without supervision and explicit instructions (Mitchell, 1997). ML is mainly used to draw inferences from data and respond to input variables. The theoretical construct of ML methods may provide better predictive performance than other

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conventional statistical models (Kaplan & Schold, 2009). Machine learning has evolved over recent years and has been widely used to tackle various applications in different fields, such as medical diagnosis (Kononenko, 2001). It is employed to construct predictive models for diagnosing various diseases. Prediction of kidney graft failure is a complicated task due to the complex nature of the human immune system. In addition, the presence of two or more medical conditions in a patient simultaneously makes the prediction more challenging. Senanayake et al. have recently conducted a systematic review and presented a number of papers studying machine learning methods in predicting kidney graft failure (Senanayake et al., 2019). The results of their work indicate the superiority of the ML-based methods in clinical decision-making, specifically in the prediction of kidney transplantation.

Imbalanced data is one of the issues which might occur in different machine learning applications such as text classification, fraud detection, and disease diagnosis. In kidney graft failure prediction, imbalanced data is a common phenomenon. Badrouchi et al. studied the prediction of long-term graft survival in an imbalanced dataset (Badrouchi et al., 2021). Imbalanced data is a situation in which the number of instances in one class is much larger than that of the other classes. In binary classification, the class that includes more patterns is called the majority class, and the one with the fewer records is referred to as the minority class. The theoretical concept of traditional machine learning classifiers is generally suitable for classifying balanced data, and applying these classifiers to imbalanced data might result in poor predictive performance and lower the algorithm's efficiency (W. Zheng & Jin, 2020). For instance, imagine a patient dataset that has 500 instances. The number of patients with a specific disease equals 10 out of 500 records. In this case, the classifier or the data distribution should be improved; otherwise, the algorithm will likely classify all instances as the majority class (individuals with no disease). Early and timely diagnosis of diseases could significantly decrease the mortality rate. In an imbalanced data distribution, when the classifier tends to label most records as majority class, optimal and timely detection and treatment are probably not applicable. Therefore patients' lives might be endangered. Therefore, it is essential to develop methods and algorithms to handle imbalanced data in ML applications.

In this study, we propose a machine learning framework for predicting kidney graft failure with an imbalanced dataset. Our main novelty in this paper is to develop a new clustering-based method using Red Deer Algorithm (RDA) to propose a novel three-stage under-sampling algorithm. Zheng et al. proposed a three-stage under-sampling algorithm to handle imbalanced datasets. It consists of denoising, fuzzy c-means clustering, and representative sample selection (M. Zheng et al., 2021). Fuzzy c-means has a poor performance on datasets with unequally distributed clusters. Variation in the shape and size of the clusters may also impact the result of fuzzy c-means clustering. It is also sensitive to noise and outliers (Memon & Lee, 2017). In addition, it can manage only one criterion for improving the obtained cluster centers. Hence, in this article, the same strategy as (M. Zheng et al., 2021) is employed, but a new clustering method using RDA is used for the second phase instead of fuzzy c-means. In the proposed clustering algorithm used in the second phase, we first use a grid-based clustering algorithm to partition the data space into different regions. It is compatible with datasets that have clusters with different densities because it considers data distribution when partitioning data. Then, RDA is used to perform a global search on obtained regions to achieve cluster centers. Due to the strong search ability and being suitable for high-dimensional data, we use a modified version of RDA to optimize the cluster validity index in order to achieve superior cluster centers. These cluster centers are then used to address the class imbalance problem and reduce data dimensionality. The quality of cluster centers can determine the efficiency of the proposed undersampling method, so using an efficient optimizer, RDA, which can achieve excellent solutions, will help us to continue the following tasks better. This clustering method is less sensitive to noise and outliers due to the strong search ability of RDA. It is also capable of optimizing cluster centers regarding multiple objective functions.

The rest of this study is organized as follows. In Section 2, a literature review is presented, and in Section 3, related concepts are given. Subsequently, in Section 4, the proposed method is described, followed by the associated results in Section 5. Finally, the conclusion is presented in Section 6.

2. Literature review

As mentioned before, Senanayake et al. performed a systematic review and studied 18 papers in the field of kidney transplantation failure prediction. Based on their results, decision trees were the most popular ML classification algorithm associated with kidney graft failure prediction (Senanayake et al., 2019). Shaikhina et al. introduced a novel method that predicted early antibody-incompatible kidney transplant rejection using a decision tree (Shaikhina et al., 2019). Pre-transplant features were used to train the model on a small dataset with only 80 patients. They predicted the rejection with an accuracy of 85 %. Some key risk factors were also indicated in their work. In contrast, Yoo et al. tried to predict kidney graft rejection on a relatively large dataset, including 3,117 kidney transplant recipients (Yoo et al., 2017). They experimentally showed that ensemble models outperformed the conventional models.

Badrouchi et al. implemented Synthetic Minority Oversampling Technique (SMOTE) to oversample their imbalanced dataset (Badrouchi et al., 2021). Their dataset included pre, peri, and post kidney transplantation features. They utilized four machine learning algorithms: logistic regression, k-nearest neighbors, extreme gradient boosting, and artificial neural network. The results indicated that extreme gradient boosting method had superiority over other methods and achieved the highest area under the curve (AUC), 89.7 %.

Naqvi et al. developed a machine learning model to predict the risk of kidney transplantation failure among three different cohorts, which is within one year, within five years, and after five years following the graft (Naqvi et al., 2021). They applied a deep learning-based autoencoder for the reduction of data dimensionality. It was shown that this data dimensionality reduction significantly improved the predictive model's performance. Their model was able to predict the graft survival with the area under the curve of 82 %.

Decruyenaere et al. studied the performance of nine different machine learning approaches to predict delayed graft function after kidney transplantation (Decruyenaere et al., 2015). Their datasets included 497 patients who received a kidney from deceased donors. They also applied a feature elimination process and selected 20 features out of 55 existing ones. The results revealed that the linear support vector machine was the most appropriate model for predicting delayed graft function.

Tapak et al. attempted to identify prognostic factors related to kidney graft rejection using an artificial neural network (ANN) (Tapak et al., 2017). They compared the obtained results from ANN with logistic regression (LR). They utilized a dataset consisting of 378 patients who received kidney transplantation from 1994 to 2011 in Hamadan, Western Iran. Their results showed that recipients' age, creatinine level, cold ischemic time, and hemoglobin level at discharge were the most critical factors in diagnosing kidney graft failure. It was also indicated that ANN outperformed LR regarding the area under the curve (88 % vs 75 % for LR).

Krikov et al. considered information regarding 92,488 patient records from the US Renal Data System (Krikov et al., 2007). Their decision tree model for predicting kidney allograft survival at different time intervals (1, 3, 5, 7, and 10 years following kidney transplantation) included 31 features, including pre and post-transplant indicators. The model achieved AUC of 0.63, 0.64, 0.71, 0.82, and 0.9 for 1, 3, 5, 7, and 10 years, respectively.

As mentioned earlier, decision tree models were the most commonly used machine learning algorithm in the field of kidney graft failure

prediction. Greco et al. concluded that decision tree could be a wise choice in clinical applications (Greco et al., 2010). This is because the results obtained from the decision tree are more interpretable and allow one to understand the interactions among different features better.

Fu et al. aimed to use machine learning models to reveal gene sets that can predict the transplantation tolerance after kidney replacement (Fu et al., 2021). They used data from 63 patients to compare 14 machine learning algorithms. They concluded that Best Subset Selection (BSS) regression method was the most efficient model among other alternatives, with a sensitivity of 91.7 %.

In another study, an image processing approach using deep learning was employed to recognize normal and abnormal kidney tissues (Yi et al., 2022). They used information related to 789 patients in two independent cohorts. They showed that their model was able to recognize kidney tissues accurately. They also demonstrated that the interstitial and tubular abnormality scores were truly predictive of graft failures after one year of transplantation.

Kers et al. adopted the same strategy of using convolutional neural networks (CNNs) to build a classifier that can group kidney allograft biopsies into three categories: normal, rejection, and other diseases (Kers et al., 2022). They utilized 5,844 digital images from 1,984 patients. For validation, they used a 3-fold validation in addition to external real-world data in terms of area under the receiver operating characteristic curve. Their model classified biopsies as normal with AUC of 0.87, diseased with AUC of 0.87, and diseased kidneys into rejection with AUC of 0.75.

Finally, like Senanayake, Sekercioglu et al. performed a scoping review for the use of machine learning in predicting long-term kidney allograft survival (Sekercioglu et al., 2021). They found that the decision tree was the most commonly used algorithm in predicting kidney transplantation rejection, followed by artificial neural networks.

3. Related concepts description

This section presents all the concepts used in this study in the following sub-sections.

3.1. Class imbalance

Class imbalance is mainly referred to as the circumstance in which the number of instances in one class is far more than that of other classes. Imbalanced data could lead to misclassification of data points. In other words, the model tends to classify all the records as the majority class. Hence, this problem should be addressed carefully to achieve more accurate results. There are usually-two main approaches to handling the class imbalance problem: data-level and algorithm-level methods (M. Zheng et al., 2021).

Data-level methods usually focus on resampling algorithms to alter the original distribution of the initial dataset so that the final data is almost balanced. Algorithm-level methods generally adopt strategies to emphasize the correct classification of the minority class samples.

Data-level methods used in this work are usually categorized into three types: oversampling, undersampling, and hybrid-sampling methods. Oversampling methods try to add some synthetic data to smooth the distribution between two classes. On the other hand, undersampling methods aim to balance the distribution by removing some non-informative instances from the majority class. Hybrid-sampling methods use both strategies simultaneously. Several undersampling methods have been proposed so far. In this study, a cluster-based approach is employed to address the class imbalance issue.

In this paper, the prediction of kidney graft failure is concerned, which is a binary classification task due to the outcome of the transplantation. In this case, there are two different classes, one of which is fewer in number. To handle the class imbalance problem, we use a clustering-based undersampling method. In some scenarios where multi-class classification is concerned, the utilization of our undersampling method could not be possible since the number of data in some unrelated classes might not be enough. In these situations, if we employ an undersampling approach, the number of input data in the final dataset is likely to be insufficient to run a predictive model. Therefore, other approaches such as oversampling or cost-sensitive learning are preferred.

It is essential to address incomplete and missing data before employing any methods that try to make the data distribution balanced. Deletion and replacement are two main approaches to dealing with missing data. The former approach is mainly used where the number of samples is large. Thus, in our case, if the missing data exists in the majority class or the minority class under the condition that we have a large number of samples in the associated class, the choice is to remove the incomplete data. However, when it comes to situations where the number of records is insufficient, deletion is not a wise choice, and the latter method, replacement, might be efficient. There are various methods for replacing missing data, such as mean substitution, regression imputation, last observation, etc. In this paper, due to our dataset's limited number of samples, we employed the mean substitution method to handle missing data.

3.2. CLIQUE algorithm

Clustering is a method that tries to divide data into similar groups in a way that objects in the same group are similar to each other and different from other groups (Berry & Linoff, 2000). Clustering can be classified into different categories such as hierarchical, partitional, graph, grid, and density-based clustering methods, each of which can be utilized in a specific application. Grid-based clustering approaches are most commonly used when the data dimension is large. It is also used when data points are not equally distributed through the data space. In these cases, grid-based algorithms show better clustering results.

Clustering In QUEst (CLIQUE), proposed by Agrawal et al., is a combination of grid and density-based methods suitable for clustering large and multi-dimensional datasets (Agrawal et al., 1998). This algorithm can automatically find clusters in associated sub-spaces of high-dimensional data. In addition, it requires a few user-specified parameters, so user interventions are not much needed. Each cluster is represented by a disjunctive normal form (DNF) expression in this method. The steps of CLIQUE algorithm are given in Algorithm 1.

Algorithm 1 CLIQUE algorithm

1. Select a suitable value for m (interval length) and τ (sensitivity threshold as a dense unit criteria).
2. Divide each dimension into m equal intervals to construct the grid structure.
3. Adopt a bottom-up strategy to find dense units. Start from the lowest dimension and generate subsequent dens units in the following levels.
4. Merge connected dense units to form a cluster.
5. Describe each cluster by a DNF expression.

3.3. Metaheuristics

A metaheuristic is a high-level strategy to find sufficiently good solutions in mathematical optimization. These methods can provide satisfying solutions, especially when the data dimension is large. Some of these heuristics algorithms are recently developed and inspired by nature, such as Red Deer algorithm (Fathollahi-Fard et al., 2020), Poplar Optimization Algorithm (Chen et al., 2022), Human Felicity Algorithm (Verij Kazemi & Fazeli Veysari, 2022), and Dwarf Mongoose Optimization Algorithm (Agushaka et al., 2022).

3.3.1. Red Deer algorithm

Fathollahi-Fard et al. have recently proposed a new nature-inspired

metaheuristic algorithm called Red Deer Algorithm (RDA) (Fathollahi-Fard et al., 2020). They experimentally showed that their proposed algorithm outperformed others in real-world problems. It is inspired by the mating behavior of Scottish red deer in a breeding season. Like other metaheuristics, there are candidates called Red Deer (RD), seeking to find optimal solutions. RDA's three main steps include roaring, fighting, and mating.

At first, RDs are categorized into male and hind based on their objective function values. A number of best solutions are marked as male RD. In the roaring process, each male RD tries to expand its territory. The new solution obtained by the roaring process is calculated by Equation (1). Then mark a percent of best male RD as commanders and the rest as stags. Commanders fight with some stags randomly and generate two new solutions calculated by Equations (2) and (3). In the breeding season, when a commander and stage approach each other for fighting, one will win, and the other will be the loser. Therefore, since our proposed clustering algorithm is a bio-inspired one, in order to show the connection between the proposed formulas and the nature of male RDs, two different solutions are generated using Equations (2) and (3). One of them, the one with better value, represents the winner and another the loser. These two solutions are used in each iteration, and the better solution will take the place of the commander. Finally, each commander forms a harem consisting of several hinds. A commander mates with some hinds in its harem and a number of hinds in other harems. The offspring of this mating process can be calculated by Equation (4). The pseudo-code of RDA is presented in Algorithm 2.

$$male_{new} = \begin{Bmatrix} male_{old} + (a_1 \times ((UB - LB) \times a_2) + LB), if a_3 \geq 0.5 \\ male_{old} - (a_1 \times ((UB - LB) \times a_2) + LB), if a_3 < 0.5 \end{Bmatrix}, \quad (1)$$

$$New_1 = \frac{(commander + stag)}{2} + (b_1 \times (((UB - LB) \times b_2) + LB)) \quad (2)$$

$$New_2 = \frac{(commander + stag)}{2} - (b_1 \times (((UB - LB) \times b_2) + LB)) \quad (3)$$

$$Offs = \frac{(maleRD + Hind)}{2} + ((UB - LB) \times c), \quad (4)$$

where UB and LB are the upper and lower bound of the search space, respectively. a_1, a_2, a_3, b_1, b_2, and c are the samples from a uniform distribution between zero and one.

Algorithm 2 Red deer algorithm

generate a population of RDs of size n
calculate the fitness function for all candidates
mark N_{male} of the best RDs as male RDs and the rest of them as hinds
repeat
  for each male RD **do**
    roar male RDs using Equation (1)
    if the fitness function of the new solution is better **then**
      replace the new solution with the previous one
    end if
  end for
  sort male RDs according to their fitness function
  choose γ percent of the best male RDs as commanders and the rest of them as stags
  for each commander **do**
    fight between commanders and stags randomly using Equations (2) and (3)
    if the objective function of new solutions is better **then**
      update the position of the commander and stag
    end if
  form harems and dedicate a commander to each harem
  mate commander with several hinds of his harem randomly using Equation (4)
  select another harem randomly
  mate the commander with some of the hinds in the selected harem using Equation (4)
  end for
  for each stag
    find the nearest hind to the stag regarding the distance function
mate the stag with the nearest hind using Equation (4)
  end for
  choose the next generation utilizing the roulette wheel selection mechanism
until the termination criterion is met
return the best solution

(continued on next column)

(continued)

3.4. Classification algorithms

This sub-section describes five machine learning algorithms used in this study.

3.4.1. Support vector machine

Support vector machine attempts to find a hyper-plane maximizing the distance between two nearest instances from two separate classes. SVM tries to minimize the following problem.

$$f(x) = sign(<w^*, X> + b), \quad (5)$$

$$w^* = \sum_{i=1}^{l} \alpha_i y_i X_i, \quad (6)$$

$$b^* = -\frac{1}{2} < w^*, X_r + X_s >, \quad (7)$$

where X_r and X_s are the support vectors, y_i is the label of the object X_i, and α_i is the Lagrange multiplier. SVM is known for its good generalization capability and has been widely used in different applications since it was introduced. Gu et al. proposed a novel support vector machine algorithm, named F-SVM, using the fly optimization algorithm for weight optimization (Gu et al., 2021). They showed that FOA-F-SVM could find better model parameters, resulting in lower computational cost and higher classification performance.

3.4.2. Artificial neural network

An Artificial Neural Network (ANN) classifier has at least three layers. One input layer, one output layer, and one or more hidden layers between input and output layers. Each layer consists of several nodes, and each node in each layer, except the input layer, is a neuron with its activation function. ANN uses the backpropagation strategy for training. ANN can classify data that are not linearly separable. It tries to optimize connection weights such that the error rate is minimized. Gradient boosting is usually used to optimize the connection weights.

3.4.3. K-nearest neighbor

In a K-nearest neighbor (KNN) classification algorithm, data is classified using the label of its K nearest data points. It simply calculates the distance (usually Euclidean distance) between an instance and all other data points. Then K closest instances are selected, and the most repeated class in these instances is identified as the output class. The choice of parameter K can determine the quality of the final prediction. Although the larger values of K usually lead to a less sensitive model to noise, it often causes less distinct boundaries between classes. A new variant of KNN algorithm, called KNNV, has been recently employed to classify COVID-19 patients. This variant has shown better performance in handling incomplete and heterogeneous data. Detailed descriptions can be found in Hamed's paper (Hamed et al., 2021).

3.4.4. Decision tree

According to the simplicity and interpretability of decision trees (DT), these are among the most popular and commonly used algorithms for classification tasks. In a decision tree algorithm, leaves represent the outcome class, while branches show the input features. A decision tree classifier starts with a feature and splits data into two subsets regarding a separation criterion. This process continues until a number of instances in a branch have the same label, and this label is presented as a leaf, and the branch will no longer be processed. There is a term called pruning in DT, which can reduce the size of the tree and decrease the computational

burden by removing the branches that do not have much effect on the classification model.

3.4.5. Ensemble models

A typical ensemble classification algorithm aims to build a model consisting of several weak classifiers. Then, a voting scheme is adopted to integrate all the outcome labels from weak classifiers. Subsequently, the final decision about the label of the test data is made according to the outcome of the voting scheme. Several approaches to building an ensemble model include bagging, AdaBoost, random forest, and boosting (Dong et al., 2019).

4. Proposed approach

Data preparation methods such as data transformation are one of the primary steps in machine learning models, and they could lead to more qualified data (Zhang et al., 2010). Hence, the model outcome is probably more precise. In this work, before applying the main algorithm, our kidney transplantation dataset is first normalized to avoid the side effects originating from varying scales among features. The proposed approach is described in detail in the following sub-sections.

4.1. Undersampling

Undersampling is one of the approaches to address the class imbalance problem to obtain accurate and correct results. Clustering-based undersampling approaches can retain the total distribution of the majority class. We propose a clustering-based undersampling method consisting of three main stages: denoising, RDA clustering, and sample selection. The proposed method can significantly reduce the risk of removing important and informative data in the majority class. Since noise in the imbalanced dataset might impact the clustering phase, adding a denoising phase before clustering could be helpful (M. Zheng et al., 2021). Finally, the most representative and informative data points are selected in the last phase. The steps of the proposed algorithm are depicted in Fig. 1.

4.1.1. Phase 1: Denoising

In this phase, we attempt to remove a number of data points in the majority class that are noisy, boundary, or redundant. By decreasing the dataset size, the denoising phase can lessen the computational burden of the following stages. It also guarantees the efficiency of phases two and three, which can be quickly impacted by noisy data. In this work, one-sided selection algorithm (OSS) (Kubat & Matwin, 1997) is utilized to eliminate noise, borderline, or redundant data from majority class samples. Each of these terms is defined as follows.

- Noisy samples: Noisy data in majority class samples are referred to as data points surrounded by the minority class samples.
- Borderline samples: Borderline samples in the majority class are placed near the majority and minority class boundaries. These samples might affect the model's learning step and lead to the misclassification of the test dataset.
- Redundant samples: A redundant sample in the majority class is surrounded by the samples that belong to the majority class.

There is a term called Tomek link in the OSS algorithm, defined as Equation (8).

$$dist(a,c) \leq dist(a,b) or dist(b,c) \leq dist(a,b) \tag{8}$$

where a and b are samples from minority and majority classes, respectively. It is said that a and b form a Tomek link if no third sample like c can satisfy Equation (8). Note that the *dist*() function refers to the Euclidean distance function.

Using the denoising phase and OSS algorithm does not balance the training dataset since it eliminates only a few majority class samples. Hence, it is still needed to employ other phases to obtain a balanced dataset. The steps of OSS algorithm are described in Algorithm 3.

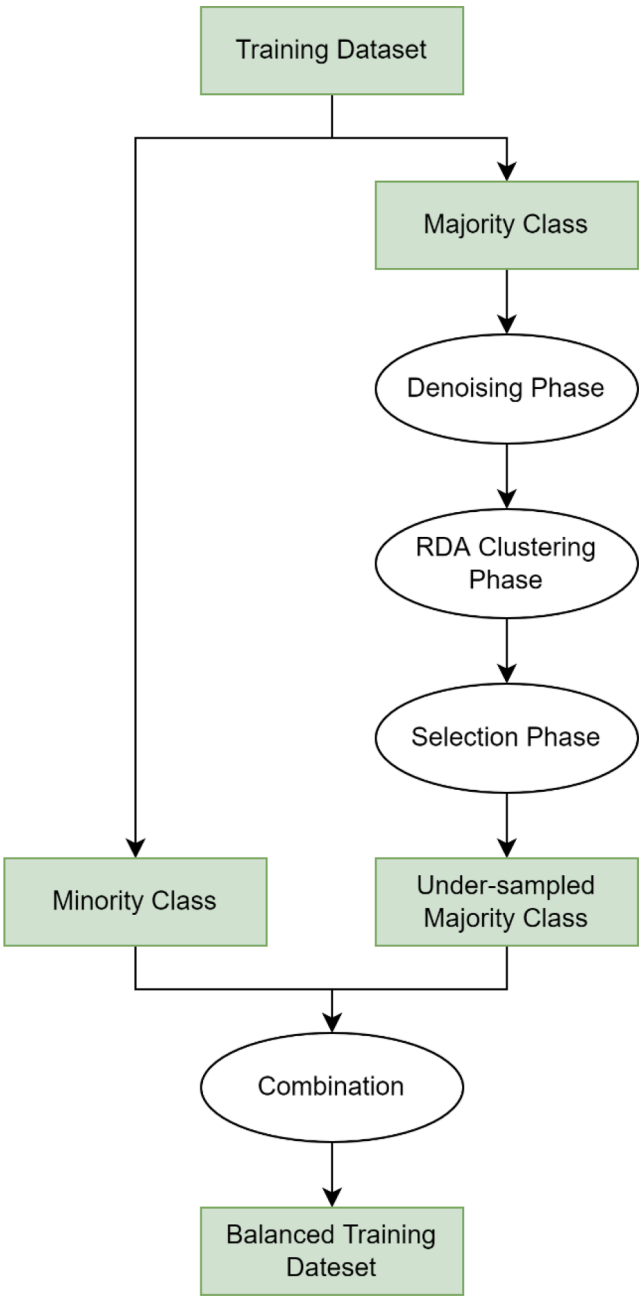


Fig. 1. Proposed approach flowchart.

Algorithm 3 OSS algorithm

divide the training dataset into majority class objects (N_{maj}) and minority class objects (N_{min})
put all samples of N_{maj} and N_{min} into one set and name it S
form another set (C) and add all N_{min} samples into it
randomly select a sample from N_{maj} and add it to C
repeat
  employ K nearest neighbor (KNN) algorithm to classify S using samples of C as the training set
  add misclassified samples to C
until C is a consistent subset of S
eliminate all N_{maj} that form Tomek links, calculated in Equation (8), and all N_{min} samples from C
return new majority class samples

4.1.2. Phase 2: RDA clustering

Clustering is an approach that aims to group a set of data points so that objects in a group, called a cluster, are as similar as possible to each other and different from other groups. Using clustering in this phase can significantly decrease the possibility of removing useful and informative majority class samples. It can also keep the overall distribution of original majority class samples. There are various approaches for clustering task. *K*-means and fuzzy *c*-means are two of the conventional clustering methods employed in most cases. Although they are easy to use and might perform well in some cases, there are several flaws and disadvantages in utilizing these methods. For example, *K*-means algorithm may trap in the local optimum, and its final solution is highly dependent on the quality of the initial solution. Fuzzy *c*-means is sensitive to noisy data, and it does not provide reasonable solutions when the data is unequally distributed. In addition, *K*-means and fuzzy *c*-means might not perform well when the number of dimensions or clusters is large. They are also capable of handling only one objective function. To overcome the problems mentioned above, we proposed a two-stage hybrid method using the RDA and CLIQUE algorithm for clustering. Due to the strong search ability of metaheuristics, they usually show good performance in multi-dimensional datasets and mainly converge to the near-global optimum (Ahmadi et al., 2010). They can also manage multiple objective functions simultaneously. In the proposed method, we first partition data space into separate regions. Then, RDA is used to traverse each division to find the cluster center associated with each division. Each RD in a region is responsible for finding the cluster center of that region.

In the first stage of the proposed clustering algorithm, we employ a grid-based clustering approach to divide search space into different regions. CIQUE algorithm is used in this stage because it can automatically find clusters in sub-spaces, and it is suitable for high-dimensional data. It also needs a few user-specified parameters. In addition, each cluster in the CLIQUE algorithm is described by a DNF expression, which can be used in the next stage to find optimal cluster centers. Since, in the proposed clustering algorithm, each candidate is responsible for finding the cluster center of its region, we define the upper and lower bounds in a way that no candidate exceeds its region. By doing so, the candidates are likely to find the best location of their cluster centers faster. Otherwise, the candidates could search in other regions that might not be a good place for their cluster center.

In the second stage, RDA is utilized to efficiently search each region to find the best cluster center according to the associated objective function, which usually is a cluster validity index such as compactness and separation. More details about the objective functions are given in Section 5. These cluster centers obtained during the optimization process are then used in the undersampling and data dimensionality reduction method. RDs in different regions cooperate to converge to a near-optimum solution. We also suggest that a new candidate participates in the roaring step. It is defined as Equation (9). We refer to this new candidate as mean RD.

$$MeanRD = (X_1, \cdots, X_K), X_k = \frac{1}{n_k}\sum_{j=1}^{n_k} d_j^k, k = 1, \cdots, K, \tag{9}$$

where n_k is the number of objects in cluster k and d_j^k represents the object j in cluster k. The steps of the proposed clustering algorithm using CLIQUE and RDA are presented in Algorithm 4.

Algorithm 4 Proposed clustering algorithm

Stage 1: Partition the data space into separate regions $r^1, \cdots, r^k$
  Utilize the CLIQUE algorithm
  Describe each region by a DNF expression
Stage 2: Adopt the following search strategy until finding the optimal solutions
  generate a population of RDs of size n
  calculate the fitness function for all RDs
  mark N_{male} of the best candidates as male RDs and the rest of them as hinds

(*continued on next column*)

(*continued*)

Algorithm 4 Proposed clustering algorithm

repeat
  for each male RD **do**
    roar male RDs using Equation (1)
    form the mean RD using Equation (9)
    select the best solution among mean RD, roared RD, and current RD
    replace the selected solution with the current RD
  end for
  sort male RDs according to the fitness function
  choose γ percent of the best male RDs as commanders and the rest of them as stags
  for each male commander **do**
    fight between male commanders and stags randomly using Equations (2) and (3)
    if the objective function of new solutions is better **then**
      update the position of the commander and stag
    end if
    form harems and dedicate a commander to each harem
    mate the commander with several hinds of his harem randomly using Equation (4)
    select another harem randomly
    mate the commander with some of the hinds in the selected harem using Equation (4)
  end for
  for each stag
    locate the nearest hind to the stag regarding the distance function
    mate the stag with the nearest hind using Equation (4)
  end for
  choose the next generation utilizing the roulette wheel selection mechanism
until the termination criterion is met
return the best solution

The number of clusters should be specified before applying the clustering algorithm. There are two main strategies for determining the number of clusters: expert opinion and similarity measures (B. Zheng et al., 2014). In this work, two similarity measures, Equations (10) and (11), are adopted to determine the optimal number of clusters.

$$v_{avg} = \frac{\sum_{k=1}^{K}\sum_{i=1}^{n_k}\sqrt{\sum_{j=1}^{d}\left(x_{ij}^k - x_j^k\right)^2}}{\sum_{k=1}^{K} n_k} \tag{10}$$

$$v_{min} = \min\left[\sqrt{\sum_{j=1}^{d}\left(x_j^{k_1} - x_j^{k_2}\right)}\right], \forall k_1 \neq k_2 \tag{11}$$

where K is the number of clusters, n_k denotes the number of data points in cluster k, d is the number of dimensions in the dataset, x_{ij}^k represents feature j of data point i in cluster k, and x_j^k is feature j of cluster center k. v_{avg} calculate the average distance between all data points and their associated cluster centers. v_{min} shows the minimum distance between two cluster centers for all cluster centers. To obtain solutions so that objects in a cluster are as much as compact and far from other groups, we tend to minimize v_{avg} and maximize v_{min}. Therefore, we define an index calculated by Equation (12).

$$K^* = arg_k \min \frac{d_{avg}}{d_{min}} \tag{12}$$

By minimizing the fraction in Equation (12), one can find the optimum number of clusters.

4.1.3. Phase 3: Sample selection

Since representative sample selection is an essential process in undersampling methods, and it can determine the results quality of the predictive model, this phase is the most determinative and critical phase, which should be well addressed. All samples in a cluster are not equally important, so this stage aims to monitor samples in each cluster and filter out the not very informative samples. The representative sample selection in this phase is based on the max–min concept. There are several terms related to this phase which are defined as follows.

- Sample size denotes the number of samples in each cluster, calculated by Equation (13). In this Equation, α is a user-specified parameter, usually between zero and two.

$$MinPts_k = \alpha\left(int\left(\frac{N_{maj}}{N_{min}}\right)\right), \alpha \in (0,2], k = 1, \cdots, K \tag{13}$$

- Neighborhood radius: This term is represented by R and defined as Equation (14). β is a user-specified parameter, usually between zero and two. X_i^k denotes the ith object in cluster k. n_k is the number of data points in cluster k, and $dist()$ refers to the Euclidean distance function.

$$R_k = \beta\left(\frac{\sum_{i=1}^{n_k}\sum_{j=1}^{n_k} dist\left(X_i^k, X_j^k\right)}{n_k^2}\right), \beta \in (0,2], \quad k = 1, \ldots, K \tag{14}$$

- Representative object set: It is denoted by H, which can be calculated by Equation (15). In this Equation, $C_r(x)$ represents the number of samples in the R-neighborhood of x.

$$H_k = \{x | C_r(x) \geq MinPts_k, x \in n_k\}, k = 1, \cdots, K \tag{15}$$

- Screened samples: Scr denotes the number of samples that should be monitored, defined as Equation (16).

$$Scr_k = int((n_k/N_{maj})^* N_{\min}), k = 1, \cdots, K \tag{16}$$

- Selected samples: In each cluster, samples that are finally selected for the majority class are denoted by U_k. This set contains all selected samples from cluster k.

By employing the sample selection algorithm, the final ratio of minority to majority classes in the dataset is close to one. Each cluster is traversed in this algorithm to obtain a final balanced dataset. The steps of the sample selection phase are presented in Algorithm 5.

Algorithm 5 Sample selection algorithm

for cluster k ($k \in \{1, \ldots, K\}$) **do**
  calculate Scr_k using Equation (16)
  calculate $MinPts_k$ using Equation (13)
  calculate R_k using Equation (14)
  for each sample in cluster k, **do**
    if the sample contains at least $MinPts_k$ in its R-neighborhood, **then**
      tag it as a representative sample and add it to set H_k
    end if
  end for
  find the representative sample in H_k that has the most samples in its R-neighborhood and name it p_k^1
  add p_k^1 to U_k and remove it from H_k
  find the farthest sample in H_k to p_k^1 and name it p_k^2
  add p_k^2 to U_k and remove it from H_k
  for $i \in \{3, \ldots, Scr_k\}$ **do**
    for each member in H_k **do**
      calculate the Euclidean distance between the selected member of H_k and all members in U_k
      save the one that has the minimum distance
    end for
    between all saved distances, select the one that has the maximum value
    name the associated data point to the maximum value as p_k^i
    add p_k^i to U_k and remove it from H_k
    if $H_k = \varnothing$ **then**
      break
    end if
  end for
union all $U_k \forall k \in \{1, \cdots, K\}$ and tag it as the balanced majority class
return balanced majority class samples

4.2. Feature selection and extraction

High dimensional data might impact the accuracy of classification algorithms and reduce the quality of such predictive models. Therefore, feature selection and extraction are essential in machine learning algorithms. There are various approaches for feature selection and extraction. Zheng et al. proposed a novel approach for feature extraction in breast cancer recognition (B. Zheng et al., 2014). They first utilized the K-means algorithm to cluster malignant and benign tumors independently. Subsequently, the fuzzy membership function was used to calculate the membership degree in each group for each object. Then, these membership degrees were considered new features for training a support vector machine model. Ahmadi et al. used the PSO-based clustering method instead of the K-means algorithm (Ahmadi & Afshar, 2015). They experimentally showed that their proposed method outperformed the K-means method.

In this paper, we first cluster minority and majority class samples separately. RDA-clustering algorithm, described in detail in the previous section, is employed to cluster data. Afterward, a membership degree using Equation (17) is calculated for each instance instead of a fuzzy membership function.

$$md_i^k = \frac{dist(X_i, m_k)}{\sum_{j=1}^{K} dist(X_i, m_j)}, i = 1, \cdots, N, k = 1, \cdots, K \tag{17}$$

where N is the total number of instances, K is the number of clusters, X_i is the ith instance in the dataset, and m_k is the cluster center of cluster k. Finally, md_i^k gives the membership degree of data point i in cluster k. Using Equation (17), each record obtains K new features. So after the feature extraction process, the number of features is equal to the number of clusters.

4.3. Prediction models

After extracting new features, machine learning algorithms are employed. In this work, the four most popular classification algorithms are used as prediction models, namely Support Vector Machine (SVM), Artificial Neural Network (ANN), K-Nearest Neighbor (KNN), and Decision Tree (DT).

We use the radial basis function kernel or RBF kernel in SVM model. The parameter that trades off the correct classification and the decision function margin is set to 0.1. Regarding the ANN model, it has one hidden layer with 100 neurons. Moreover, the rectified linear unit function (ReLU) is used as the activation function for the hidden layer. The stochastic gradient descent method is utilized for weight optimization. The ANN model is trained over 200 iterations with a constant learning rate of 0.001, and the momentum parameter is also set to 0.9. With regards to the KNN model, three nearest neighbors are considered to determine the test data class. Turning to the DT model, C4.5 is used to generate DT, and the Gini criterion is employed to evaluate the quality of a split. Although there is no predetermined depth limitation, each internal node should have at least two instances to be split. A sample of the DT model composition is shown in Fig. 2. In this figure, $X[i]$ represents the ith feature.

Ensemble learning approaches usually show good performance in classification tasks. Moreover, when dealing with class imbalance, their predictive abilities outperform conventional machine learning algorithms (Alves Ribeiro & Reynoso-Meza, 2020). Hence, we also employ an ensemble method by aggregating those four algorithms mentioned above. The flow chart of the proposed ensemble method is depicted in Fig. 3.

5. Experimental results

In this section, the result of applying the proposed undersampling and feature extraction method on the kidney transplantation dataset using five different predictive models, namely SVM, ANN, KNN, DT, and ensemble method, are presented. A 10-fold cross-validation method is used to evaluate the performance of different classification algorithms

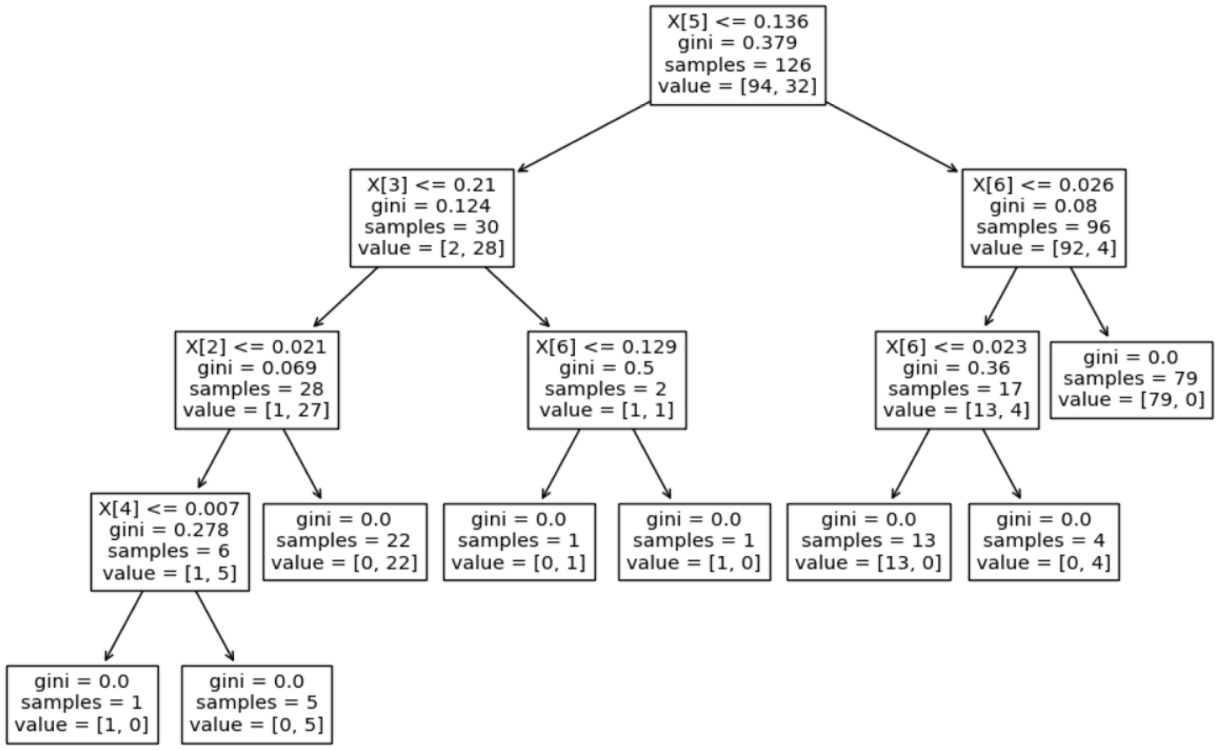


Fig. 2. A sample of a DT composition.

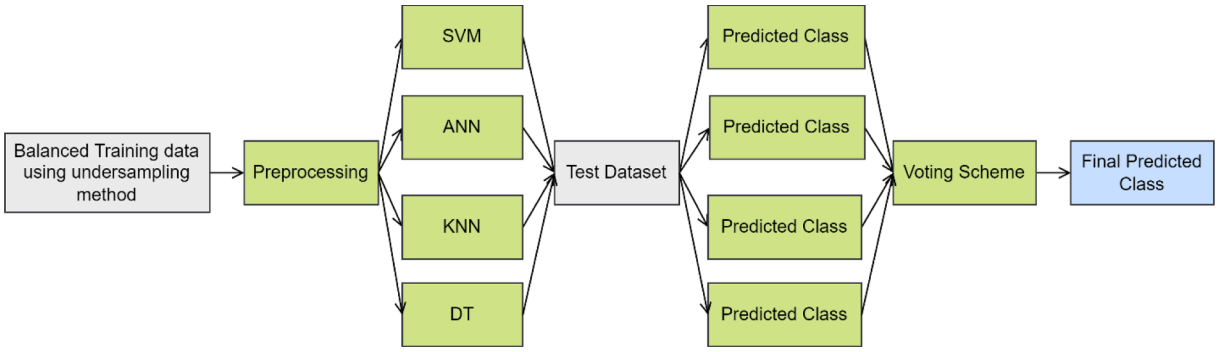


Fig. 3. Flow chart of proposed ensemble classification method.

applied to the dataset. Note that the training dataset is only undersampled, so the test data would never change, and the results are reliable. Using Python, all the machine learning algorithms were employed in the Scikit-Learn library. In this study, we have adopted a dataset collected between 1994 and 2011 in western Iran. The dataset includes 378 patients who have undergone kidney transplantation in two selected hospitals. This data has 19 features, including pre and post-transplantation variables. The last feature is the response variable (data class), defined as having chronic nonreversible graft rejection. Table 1 demonstrates the data summary. The values in Table 1 are written as number (%) or mean ± standard deviation.

As explained in Section 4, our approach to the class imbalance problem and feature extraction is based on the clustering technique. Therefore, we need to define an index to measure the efficiency of our proposed clustering algorithm. We first introduce two similarity measures, namely compactness and separation, which can be calculated by Equations (18) and (19).

$$Compactness\left(m^1, m^2, \cdots, m^k\right) = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{n_k}\sum_{i=1}^{n_k} d\left(m^k, x_i^k\right), \tag{18}$$

$$Separation\left(m^1, m^2, \cdots, m^k\right) = \frac{1}{K(K-1)}\sum_{i=1}^{K}\sum_{k=i+1}^{K} d\left(m^k, m^i\right), \tag{19}$$

where m^k is the center m of cluster k, and x_i^k is the data point i in cluster k. Clustering algorithms tend to minimize compactness and maximize separation measures. Thereby, we can define a combined validity index as calculated by Equation (20).

$$Combined\left(m^1, m^2, \cdots, m^k\right) = \alpha.Compactness\left(m^1, m^2, \cdots, m^k\right) - \beta.Separation\left(m^1, m^2, \cdots, m^k\right) \tag{20}$$

where α and β are the parameters that control the weight of each term and $\alpha + \beta = 1$.

In our proposed clustering method, the metaheuristic algorithm, called RDA, tries to optimize the objective function shown in Equation (20). The lower the cluster validity index value is, the better the clustering result is obtained. Now, this index can be used to compare different clustering algorithms. Before applying clustering methods, the optimal number of clusters is obtained using Equation (12).

Table 1
Data summary (Tapak et al., 2017).

Features	Recipient	Donor
Sex		
Male	193 (51.1)	283 (74.9)
Female	185 (58.9)	91 (24.1)
Blood group		
O	125 (33.2)	166 (44.6)
A	139 (36.9)	114 (30.6)
B	89 (23.6)	87 (23.4)
AB	24 (5.4)	5 (1.3)
Turn of transplantation		
First	361 (95.5)	
Second	16 (4.2)	
Usage of immunosuppressive drugs		
Imuran, prednisolone, cyclosporine	154 (40.8)	
CellCept, prednisolone, cyclosporine	223 (59.2)	
Post-transplantation condition		
No complication	324 (92.8)	
Acute rejection	23 (6.6)	
Hyperacute rejection	2 (0.6)	
Type of donors		
Alive donor		361 (96.3)
Deceased donor		14 (3.7)
Reason for end-stage kidney disease		
Diabetes	10 (2.7)	
Hypertension	87 (23.1)	
Glomerulonephritis	50 (13.3)	
Polycystic kidney	8 (2.1)	
Urologic disorder	7 (1.9)	
Congenital	6 (1.6)	
Unknown	208 (55.3)	
Cold ischemic time (min)	36.81 ± 5.53	
Age (year)	37.37 ± 13.64	28.08 ± 6.38
Weight (Kg)	57.22 ± 12.60	
Creatinine level at discharge (mg/dL)	1.36 ± 1.05	
Duration of hospitalization (day)	27.35 ± 10.59	
Duration of dialysis (day)	515.12 ± 587.01	
Urine volume 24 h after discharge (mL)	12,650.44 ± 8,329.75	
Hemoglobin level (mg/ dL)	10.79 ± 1.89	

In order to show the superiority of our proposed clustering algorithm over other methods, we employ the proposed method on three different datasets and compare the results with other methods and heuristics in terms of different validity indices. Two datasets are wine and glass identification data taken from UCI (Newman et al., 1998) and another Gaussian dataset represented by N4D2, meaning it has four features in a two-dimensional space. Another cluster validity index, called Turi measure, is defined by Equation (21) to compare the performance of different clustering algorithms.

$$Turi\left(m^1, m^2, \cdots, m^k\right) = \left(c \times N(2,1) + 1\right) \times \frac{a}{b} \tag{21}$$

where c is a constant value determined by the user, N(.) is Gaussian distribution with mean two and variance one. The term a is compactness, and b can be calculated by Equation (22). The clustering algorithms tend to minimize Turi's index.

$$b = min\left\{\left\|m^i - m^j\right\|\right\}, i \in [1,\ K-1], j \in [i+1,\ K] \tag{22}$$

Several clustering methods have been selected for this evaluation: *K*-means, fuzzy c-means, simple PSO, hybrid –PSO (combination with *K*-means) (Cui et al., 2005), simple RDA, Grey Wolf Optimizer (GWO) (Mirjalili et al., 2014) and the proposed method. In the first step, Gaussian data is used to compare the performance of these algorithms. Fig. 4 shows the convergence of these algorithms in terms of combined measure, presented in Equation (20), utilizing N4D2 data.

As presented in Fig. 4, applied modifications to the original RDA algorithm have led the proposed method to provide better solutions. In addition, due to the improvements in search strategy, the proposed method converges after a few iterations. It has also outperformed the other approaches.

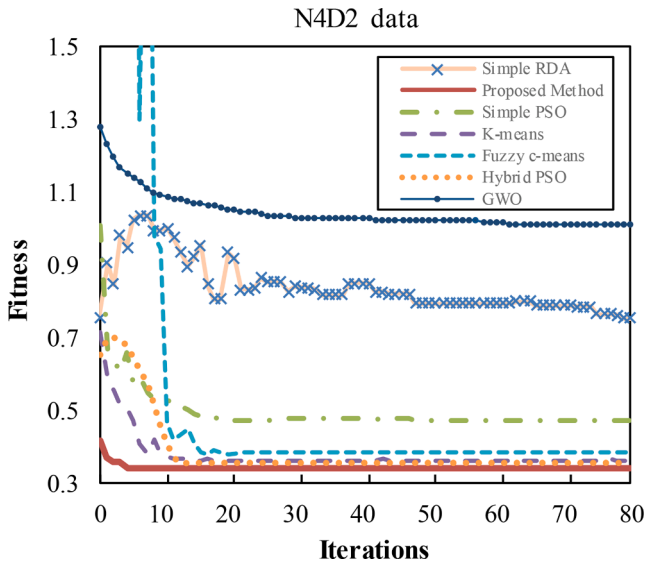


Fig. 4. Comparison of the convergence of the different algorithms using N4D2 in terms of combined measure.

As shown in Fig. 5, the proposed method outperformed the other algorithms regarding Turi's validity index in the glass identification dataset and showed a relatively good solution in the wine dataset.

After showing the superiority of the proposed clustering method over other heuristics and algorithms, it is utilized in the three-phase undersampling technique. Therefore, employing the proposed undersampling method, the final balanced dataset has 21 patients who have not experienced kidney graft rejection and 27 patients with rejection. So the balanced dataset size equals 51, and the minority to majority samples ratio is close to one. Hereafter, the class representing the patients who face failure in kidney graft is called the positive, and the other group is called the negative class.

The optimal number of clusters for each class is calculated using Equation (12). Figs. 6 and 7 show the most appropriate number of clusters for positive and negative classes based on Equation (12) under the RDA clustering approach, respectively. The number of clusters is tested between 2 and 5 to find the optimum value for *K*.

Figs. 6 and 7 show that four and five clusters minimize Equation (12) for negative and positive classes. Hence, applying the feature extraction method, the new dataset has nine features for each record. There is also a target variable that defines the class of each instance.

Different algorithms are applied to the kidney transplantation dataset, and their performance is compared using various performance evaluation metrics such as accuracy, sensitivity, specificity, *F1* score, and Area under the ROC curve (AUC). When dealing with a balanced dataset, accuracy is usually chosen as the performance evaluation metric. However, this metric is not suitable for imbalanced datasets because, in such cases, a model tends to classify all samples as majority class, resulting in high accuracy, while the model has misclassified almost all minority class samples. Therefore, other performance metrics are used to evaluate the quality of different algorithms. The confusion matrix is described first in Table 2 to help the introduction of performance metrics.

In Table 2, columns represent the predicted class, and rows show the actual class for samples in the test dataset. TP is defined as samples with a positive class in reality and predicted as positive by the model. Similarly, FN indicates that the positive samples are predicted as negative. FP shows the negative samples predicted as positive, and TN represents correctly predicted samples as negative. Accuracy, sensitivity, specificity, and *F1* score are defined as the following equations.

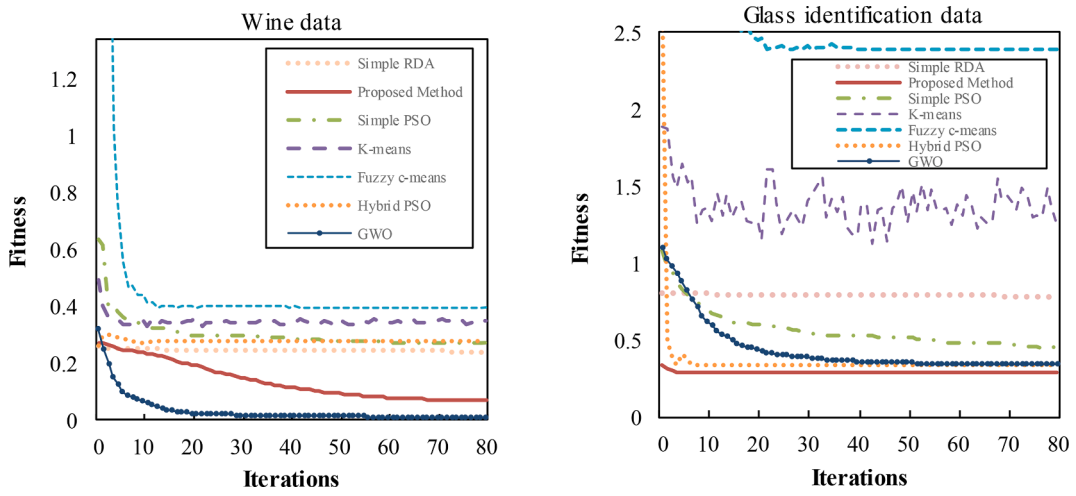


Fig. 5. Comparing the convergence of the proposed method versus other clustering approaches in terms of Turi's validity index using wine and glass identification datasets.

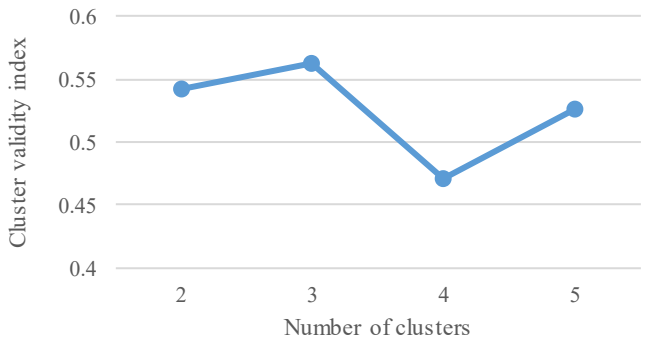


Fig. 6. Cluster validity index for negative class samples using the RDA clustering approach.



Fig. 7. Cluster validity index for positive class samples using the RDA clustering approach.

Table 2
Binary classification confusion matrix.

	Predicted positive	Predicted negative
Actual positive	True positive (TP)	False negative (FN)
Actual negative	False positive (FP)	True negative (TN)

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{23}$$

$$\text{Sensitiviy} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{24}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{25}$$

$$F1\ \text{score} = \frac{\text{TP}}{\text{TP} + \frac{1}{2}(\text{FP} + \text{FN})} \tag{26}$$

where TP is true positive, TN is true negative, FP is false positive, and FN is false negative. The larger value of AUC indicates a more efficient machine learning model. The grey region in Fig. 8 demonstrates the AUC value. In the ROC curve, sensitivity is depicted on *Y*-axis, and (1 – specificity) is plotted on *X*-axis. It can be used to see the tradeoff between sensitivity and specificity. The values of sensitivity and specificity for the optimum model equals 1. So the point (0,1) is the optimal point

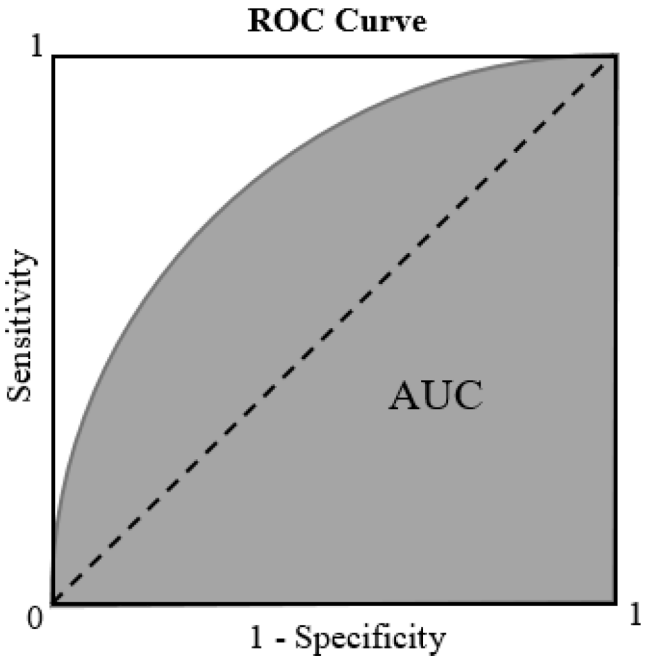


Fig. 8. ROC curve and area under the curve (AUC).

Table 3
Results.

Algorithm	Accuracy	Sensitivity	Specificity	F1 Score	AUC
SVM	0.9418 ± 0.0638	0.8833 ± 0.2363	0.9618 ± 0.0620	0.8694 ± 0.1667	0.9489 ± 0.0708
ANN	0.8632 ± 0.1231	0.5250 ± 0.4001	0.9809 ± 0.0382	0.5757 ± 0.4038	0.9509 ± 0.0574
KNN	0.9275 ± 0.0652	0.8833 ± 0.2363	0.9427 ± 0.0765	0.8463 ± 0.1613	**0.9687 ± 0.0346**
DT	**0.9643 ± 0.0479**	**0.9417 ± 0.1181**	0.9718 ± 0.0774	**0.9153 ± 0.0859**	0.9517 ± 0.0435
Ensemble	0.8918 ± 0.1188	0.6583 ± 0.4151	**0.9809 ± 0.0273**	0.6114 ± 0.3682	0.9495 ± 0.0395

The bold values highlight the best results.

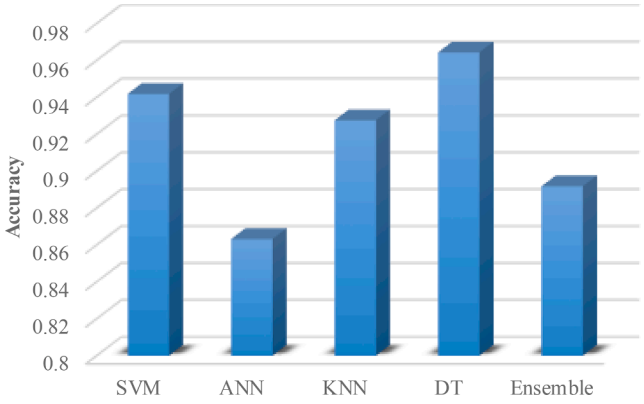


Fig. 9. Accuracy of different classification algorithms.

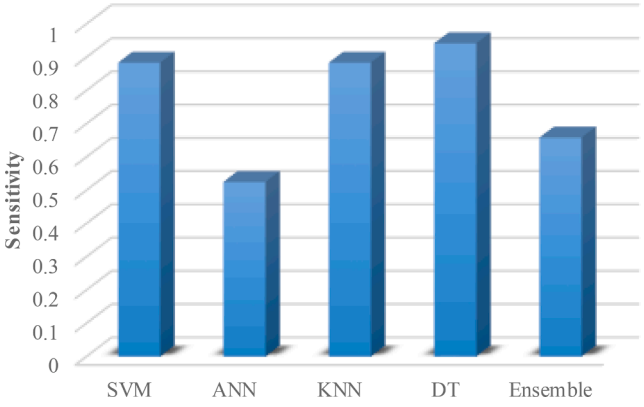


Fig. 10. Sensitivity for different classification algorithms.

in the ROC curve and represents the best classifier.

Table 3 demonstrates the results of implementing five classification methods using a 10-fold cross-validation approach to the kidney transplantation dataset. These results are also presented in Figs. 9 – 12.

As seen in Table 3, the decision tree algorithm has shown the best performance among other classification algorithms regarding three performance evaluation metrics: accuracy, sensitivity, and *F1* score. Moreover, the result obtained from a DT is more interpretable compared to other algorithms such as ANN. So it can be used as a decision support system when working with kidney graft failure prediction.

In Fig. 9, the performance of five classification algorithms in terms of the accuracy metric is compared. It shows that the decision tree model outperformed the other algorithms. After that, SVM has the best performance. The ANN model does not perform well when compared to other algorithms.

Fig. 10 demonstrates the results of machine learning algorithms

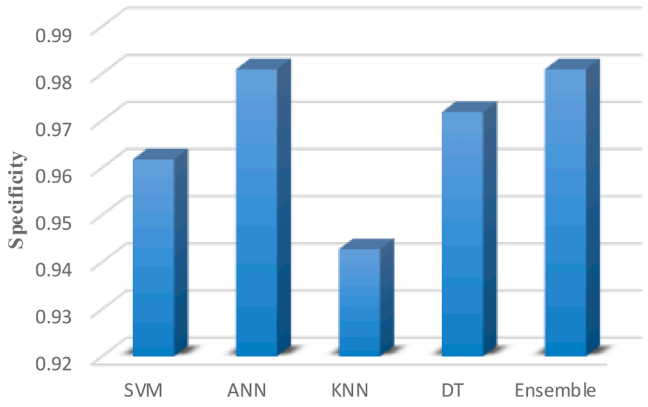


Fig. 11. Specificity for different classification algorithms.

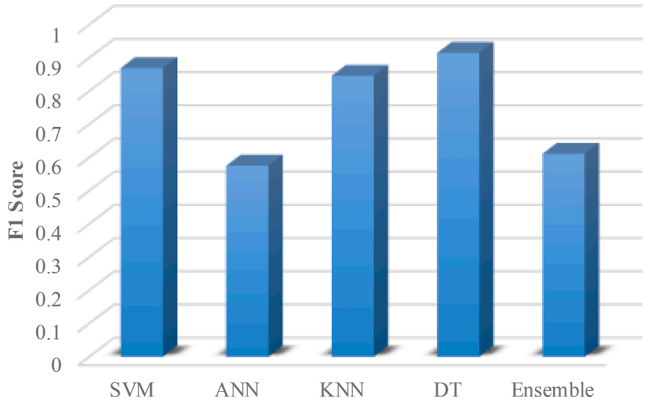


Fig. 12. F1 score for different classification algorithms.

according to the sensitivity metric. This measure is defined as how a model could classify positive class samples correctly. Since failure detection in kidney grafts is referred to as a positive class, and the correct classification of this class is critical, this measure plays a crucial role in identifying the best classification algorithm. As observed in Fig. 10, the decision tree shows superior performance in terms of sensitivity.

Specificity is the capability of a model to classify the negative class. In this work, the negative class represents not failed grafts. As mentioned before, sensitivity is more important than specificity metric due to the risk of misclassifying a positive class. From Fig. 11, it can be observed that ANN and ensemble method presented the best result in terms of specificity. The decision tree was next, and its specificity value was slightly lower.

Fig. 12 compares the performance of implemented classification algorithms in terms of *F1* score. As can be seen, the decision tree outperformed the other algorithms.

Based on the above experimental results of applying the five classification algorithms to the kidney transplantation dataset, the decision tree demonstrated relatively good performance compared to others regarding the four evaluation metrics.

As mentioned before, algorithm selection is dependent on how much risk is tolerable for each class. In other words, due to the system risk, one might select different algorithms based on sensitivity and specificity. Fig. 13 shows the ROC curve for four classification algorithms.

One can use Fig. 13 to select the best model according to system risk. In this work, having higher sensitivity is preferable to that specificity.

6. Conclusions

Over the recent decades, the end-stage renal disease has been one of the most widespread medical conditions among individuals worldwide.

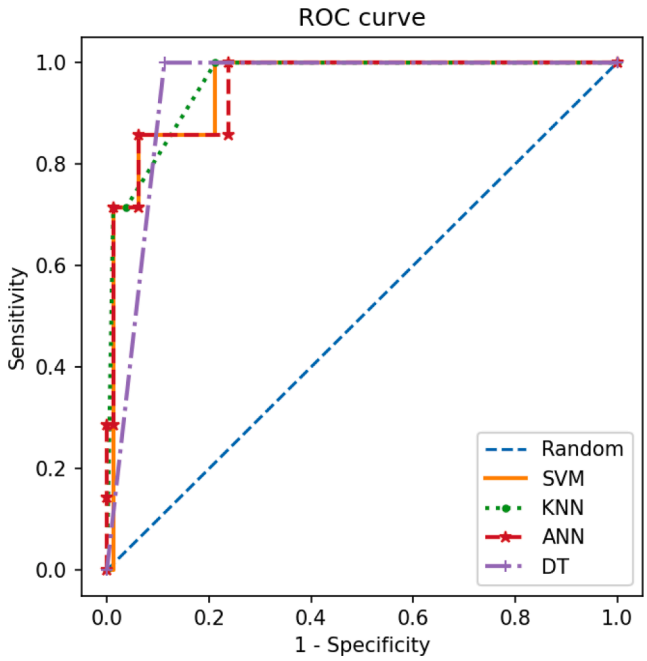


Fig. 13. ROC curve for different classification algorithms.

A patient's kidney usually needs to be transplanted for end-stage kidney diseases. Thus, the demand for kidney replacement has risen. Due to the complex nature of the human body, transplantation might fail and endanger patients' lives. Therefore, finding an appropriate donor is of great importance. Although prediction of graft failure is a complicated task, machine learning algorithms have shown superior performance in clinical decision-making, such as kidney graft rejection prediction. In disease diagnosis, the class imbalance is a common issue that should be carefully addressed. Otherwise, the model tends to label all instances as majority class samples. In this study, it was desired to predict kidney graft rejection using machine learning algorithms. We adopted a kidney graft dataset of 378 patients who had undergone kidney replacement. A three-phase clustering-based undersampling approach was introduced to address the class imbalance problem.

The undersampling method consists of denoising, RDA clustering, and sample selection phases. In the second phase, a novel metaheuristic-based clustering approach using the red deer algorithm was presented to cluster majority class samples. The results of the cluster validity index showed that the proposed clustering algorithm outperformed other conventional ones like *K*-means and fuzzy c-means. It was also compared with other metaheuristics, such as PSO and GWO, and the superiority of the proposed clustering algorithm was shown. Employing the undersampling method, the ratio of majority to minority class samples is close to one, and a classification algorithm was expected to perform well. We proposed a feature extraction method based on the RDA-based clustering method to decrease the data dimensionality. This method reduced the dimensionality of data from 19 to 9. Then five classification algorithms, namely SVM, ANN, KNN, DT, and ensemble method, were implemented to predict kidney graft failure. To evaluate the performance of predictive models, accuracy, sensitivity, specificity, *F1* score, and AUC of each algorithm were reported. The results showed that the decision tree model outperformed the other algorithms with 0.96, 0.94, 0.97, 0.95, and 0.95 for accuracy, sensitivity, specificity, *F1* score, and AUC, respectively. Moreover, the results obtained from decision trees are more understandable and interpretable. Hence, the proposed model in this study could be used as a decision support system for clinical experts to predict kidney graft failure.

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CRediT authorship contribution statement

Pourya Moghadam: Conceptualization, Methodology, Software, Writing – original draft, Validation, Formal analysis. **Abbas Ahmadi:** Supervision, Methodology, Writing – review & editing, Validation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The kidney transplantation dataset and N4D2 analyzed during the current study are available from the corresponding author on request. Glass identification and wine datasets are publicly available at the UCI machine learning repository.

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